INCH-POUND

MIL-C-82807(OS)
29 December 1989
SUPERSEDING
WS 20647
25 October 1983
(See 6.6)

MILITARY SPECIFICATION

COATING, FLAMEPROOF

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of flameproof coating, referred to herein as coating.

1.2 Classification. The flameproof coating shall be of the following types (colors), as specified in the contract (see 6.2):

Type I	Color: Flat white
Type II	Color: Flat black

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8010

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FEDERAL

FED-STD-595 Colors

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 Transportation

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402-0001.)

NAVAL SEA SYSTEMS COMMAND (CAGE Code 10001)

WR-43 Preparation of Quality Assurance Provisions (Including Classification of Characteristics)

(Copies of WR-43 are available from the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-4677.)

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NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample of coating shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

3.2 Material. The coating shall consist of a single component silicone resin coating furnished in aerosol cans.

3.3 Performance properties. After aerosol packaging, the coating shall conform to 3.3.1 through 3.3.3.

3.3.1 Color. The color of the cured coating shall be flat white (Type I) or flat black (Type II).

3.3.2 Flammability. The coating shall not burn when subjected to a temperature of 1200 to 1250°F for 40 to 60 minutes.

3.3.3 Adhesion. The cured coating shall show no gross delamination and no separation from the substrate when subjected to a dry tape test. A slight "dusting" (removal of some surface particles) on the tape's adhesive surface is acceptable.

3.4 Lot identification. Each container shall be coded on the bottom with a filling date and shall be traceable to the bulk sample requirement of 3.2 and the aerosol packaging requirement of 3.3.

3.5 Shelf life. The coating shall be stored at 60 to 90°F. The shelf life shall be 18 months from date of manufacture under this condition.

3.5.1 Temperature excursions. The maximum cumulative exposure time out of controlled storage shall be 30 days.

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3.5.2 Shelf life extension. The shelf life expiration date of an individual lot of coating may be extended for an additional 6-month period provided the coating, upon retest, meets the requirements specified in 3.3.1 through 3.3.3.

3.6 Workmanship. Workmanship shall be such as to ensure a uniform product free from contamination and other defects that would affect the coating properties.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the tests specified in 4.5.

4.4.1 Lot. A lot shall consist of all material manufactured using the same procedures and subject to inspection at one time.

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4.4.2 Sampling. A sample consisting of one kit or unit container (one 13 ounce aerosol can) shall be selected at random from each lot of material. Failure of the sample to pass any of the tests specified herein shall be cause for rejection of the lot.

4.4.3 Classification of characteristics. The characteristics verified by the inspections herein are classified as Major or Minor in accordance with WR-43. Inspections that verify Major characteristics are identified by the symbol (M). The number following the classification symbol indicates the serial number of the inspection. Inspections which are not annotated with a classification code and serial number are classified minor.

4.5 Inspection methods. The following inspection methods shall be used. Unless otherwise specified in the applicable test method, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 Visual examination. Visual examination shall be made to verify conformance to the requirements of 3.2, 3.3.1, 3.6 and section 5.

4.5.2 Flammability (M101). A 3- x 6-inch 302 corrosion resistant steel test specimen shall be cleaned, dried, and sprayed at ambient temperature with two coats of material, allowing 15 to 30 minutes between coats. Cure at ambient temperature 30 minutes minimum prior to heating in an oven maintained at 1200 to 1250°F for 40 to 60 minutes.

4.5.3 Adhesion. Two 3- x 6-inch 302 corrosion resistant steel test specimens shall be cleaned, dried, and sprayed at ambient temperature with two coats of material, allowing 15 to 30 minutes between coats. Cure at ambient temperature 30 minutes minimum, then firmly apply a 3- to 6-inch long strip of 1-inch-wide masking tape to the specimens and peel off in one abrupt motion.

4.6 Inspection of packaging. The coating containers and container markings shall be visually examined to verify compliance with Section 5.

5. PACKAGING

5.1 Packaging. Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 Level C. The coating shall be packaged to afford adequate protection against loss, contamination, deterioration, and damage during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The

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packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following:

a. Title, number, and date of this specification.

b. Manufacturer's name and address.

c. Net weight of contents.

d. Shelf life and inspection/test date for extension (see 3.5.2). In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 6 MONTHS PROVIDED THE COATING, UPON RETEST, CONFORMS TO THE COLOR, FLAMMABILITY AND ADHESION REQUIREMENTS OF MIL-C-82807".

e. Storage conditions.

f. Filling date (traceable to bulk batch and aerosol packaging).

g. Color.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The coating is intended for use as an exterior or interior finish on the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

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b. Type requested (see 1.2).

c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

d. Whether first article inspection is required (see 3.1 and 6.4).

e. Special marking, if other than as specified in 5.2.2.

6.3 Possible material. The following materials have been found to meet the requirements of this specification. This is only given for information and is not restrictive. Sperex SP-101 VHT flat white paint meets the requirements for Type I and Sperex SP-102 VHT flat black paint meets the requirements for Type II.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, or a standard production item from the contractor's current inventory (see 3.1), and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

6.6 Supersedure information. MIL-C-82807 incorporates the following engineering change proposals (ECPs) and specification change notices (SCNs):

ECPs	SCNs
SSS1050 (11/9/84)	SCN 1 (9/10/84)
MTA051 (11/30/87)	SCN 2A (8/20/87)

Preparing Activity:
NAVY-OS
(Project 8010-N283)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-C-82807	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
COATING, FLAMEPROOF

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME **Commanding Officer** **Naval Ordnance Station**	b. TELEPHONE (Include Area Code) (1) Commercial **(301)743-4358** (2) AUTOVON **364-4358**
c. ADDRESS (Include Zip Code) **Code 3730** **Indian Head, MD 20640-5000**	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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